Week 2 Assignment: Market Regime Simulator using Markov Chains

Objective

The objective of this assignment is to introduce students to the concept of Markov Chains and memoryless systems by applying probabilistic state transitions to simulate simplified real-world financial market behavior and analyze the long-run evolution of different market regimes.

Background

Financial markets can broadly be classified into three major regimes: Bull, Bear, and Sideways markets. A Bull market represents a generally upward trend in prices, a Bear market represents a declining market, while a Sideways market indicates relatively stagnant price movement with no strong upward or downward trend.

Over time, the market may transition between these regimes probabilistically. Historical observations suggest, for example, that if the market is currently in a Bull state, there is a 70% probability that it remains Bull the next day, a 20% probability that it transitions to a Bear market, and a 10% probability that it moves into a Sideways regime. Similar transition probabilities are provided for the Bear and Sideways states in the transition matrix.

Market States

We define three market states:

State Number	Market State
0	Bull
1	Bear
2	Sideways

Transition Matrix

Use the following transition matrix:

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.4 & 0.3 & 0.3 \end{bmatrix}$$

Interpretation:
Each row represents the current state and each column represents the next state.

Task 1

Part 1 :- Simulate Market Evolution

Starting from an initial state:

```
Bull
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simulate the market for:

- 500 days

At every step:

- Use the transition probabilities
- Randomly generate the next state based on probability.

Store the complete list of states in an array.

Part 2 :- Frequency Analysis

Count how many times each state occurs. And fraction of occurrence of each.
Compare these frequencies with the expected long-run behavior.

Part 3 :- Visualization

Create the following plots:

- Market state vs time
- Histogram of state frequencies
- Running probability estimates over time

Task 2

Assign daily returns to each market regime.

Market State	Daily Return
Bull	+1%
Bear	-1.2%
Sideways	+0.1%

You are given an initial capital of:

$$\text{Portfolio Value} = 1,00,000$$

Update portfolio value daily using:

$$V_{t+1} = V_t(1 + r_t)$$

where:

- V_t = portfolio value at time t
- r_t = daily return corresponding to the market state

Track your portfolio by a simple plot and show your final return on the capital.

Bonus Task

Google's PageRank algorithm is one of the most well-known real-world applications of Markov Chains.

Research briefly how PageRank models a user randomly moving from one webpage to another using transition probabilities and the Markov property.

In 1-2 paragraphs, explain in your own words:

- How the Markov property is used in PageRank
- How webpages can be viewed as states in a Markov Chain
- How steady-state probabilities determine webpage importance

You may include any relevant diagram or visualization from the internet to support your explanation..

Format

You may use these libraries

- numpy
- matplotlib
- random
- pandas

You have to submit a pdf of the google colab file, which includes the code and the results. The file should be well commented, and should properly explain your working. You are encouraged to do this on your own and use of AI tools should be limited. We are not placing a restriction but do not submit a complete AI made project. Use it only for technical help in coding. And strictly do not copy from each other.